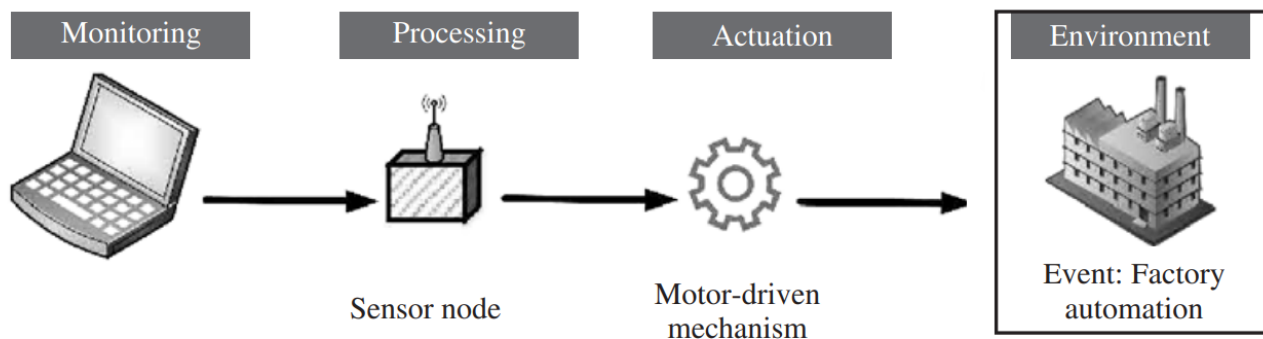

Actuators

An actuator is a mechanical or electro-mechanical device that converts energy from a control signal into mechanical motion. In simpler terms, an actuator is a device that allows controlled movement or positioning.

Actuators need a control signal and a source of energy to bring about mechanical motion. Actuators use an energy source – such as electrical, compressed air, or hydraulic pressure – and a control signal that can be manual, an automatic electronic system, a fixed mechanical system, a software-run system or a robotic control system.

Figure given below shows the outline of a simple actuation system. A remote user sends commands to a processor the processor instruct a motor controlled robotic arm to perform the commanded tasks accordingly. The processor is primarily responsible for converting the human commands into sequential machine-language command sequences, which enables the robot to move. The robotic arm finally moves the designated boxes, which was its assigned task. Soft actuators



Actuator Types

1) **Hydraulic actuator:** Hydraulic actuator is, a device that is used to change the fluid's pressure energy into mechanical force. The hydraulic actuator includes a cylinder or a fluid motor that works through hydraulic power for mechanical operation. The mechanical motion provides an output in the form of rotary, linear otherwise oscillatory motion. When liquids are almost unfeasible to compress, then a hydraulic actuator uses a large force.

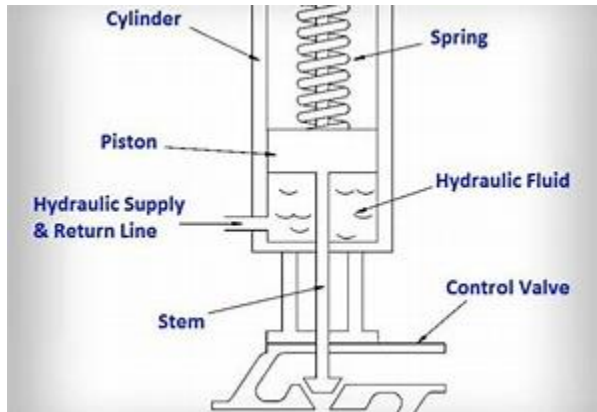


Fig: Hydraulic actuator

The **Hydraulic actuator working principle** is, it uses liquid pressure to work instead of instrument air pressure to provide a force on the diaphragm to move the valve actuator, then to the stem of the position valve. Almost all types of hydraulic actuators use a piston instead of a diaphragm for changing liquid pressure into mechanical power.

2) **Pneumatic actuators:** Pneumatic actuators are the devices used for converting pressure energy of compressed air into the mechanical energy to perform useful work. In other words, Actuators are used to perform the task of exerting the required force at the end of the stroke or used to create displacement by the movement of the piston. The pressurized air from the compressor is supplied to reservoir. The pressurized air from storage is supplied to pneumatic actuator to do work.

3) **Electric actuators:** Electric actuators typically, electric motors are used to power an electric actuator by generating mechanical torque. This generated torque is translated into the motion of a motor's shaft or for switching (as in relays). For example, actuating equipments such as solenoid valves control the flow of water in pipes in response to electrical signals. This class of actuators is considered one of the cheapest, cleanest and speedy actuator types available.

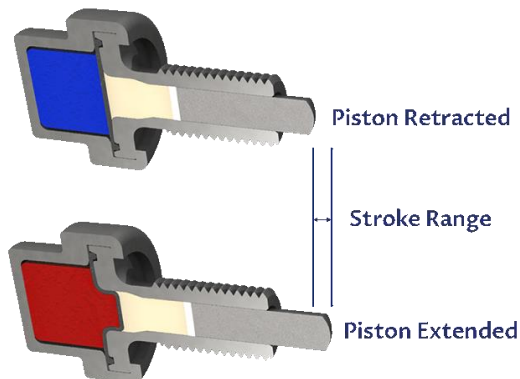
4) **Thermal actuators:** Thermal actuators convert thermal energy into kinetic energy, or motion. A typical thermostatic actuator consists of a temperature-sensing material sealed within a housing by a diaphragm that pushes against a plug to move a piston within a guide

Thermal actuators produce a linear movement – or stroke – as a result of the expansion and contraction of a thermally sensitive material within them.

A thermostatic actuator acts as a valve actuator to open or close a valve based on a temperature setpoint to, for example, continuously regulate the flow of hot water to a radiator heating a room.

Integrating temperature sensing and actuation into a single device, [thermal actuators](#) are commonly used in applications such as temperature control, fluid mixing and diverting, freeze protection and high-temperature shutoffs.

“Cold Position” - Solid State



T “Hot Position” - Liquid State

Fig: Thermal actuator

5) **Magnetic actuators:** Magnetic actuators make use of magnetic effects to generate mechanical force, a rotary or linear motion, continuous or limited motion. Magnetic actuators are found in a wide range of HVAC (Heating, Ventilation, and Air conditioning) applications to shut off or mix fluids and steam.

HVAC is the technology of indoor and vehicular environmental comfort. It aims to provide thermal comfort and acceptable indoor quality. The design of HVAC systems is a domain of mechanical engineering based on the principles of thermodynamics, fluid mechanics, and heat transfer. 6)

Mechanical actuators: In mechanical actuation, the rotary motion of the actuator is converted into linear motion to execute some movement. The use of gears, rails, pulleys, chains, and other devices are necessary for these actuators to operate. These actuators can be easily used in conjunction with pneumatic, hydraulic, or electrical actuators. They can also work in a standalone mode. The best example of a mechanical actuator is a rack and pinion mechanism. Following fig shows such an example of a rack and pinion mechanism

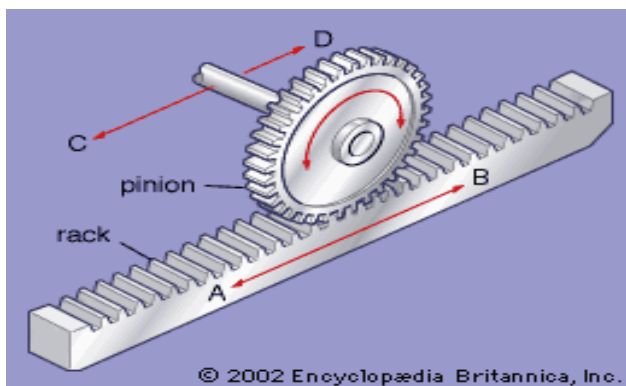


Fig: Rack and pinion mechanism

7) **Soft actuators:** Soft actuators (e.g., polymer-based) consists of elastomeric polymers that are used as embedded fixtures in flexible materials such as cloth, paper, fiber, particles, and others. The conversion of molecular level microscopic changes into tangible macroscopic deformations is the primary working principle of this class of actuators. These actuators have a high stake in modern-day robotics. They are designed to handle fragile objects such as agricultural fruit harvesting, or performing precise operations like manipulating the internal organs during robot-assisted surgeries.

8) **Shape memory polymers:** Shape memory polymers (SMP) are considered as smart materials that respond to some external stimulus by changing their shape, and then revert to their original shape once the affecting stimulus is removed. Modern-day SMPs have been designed to respond to a wide range of stimuli such as pH changes, heat changes, light intensity, and frequency changes, magnetic changes, and others.

Photopolymer/light-activated polymers (LAP) are a particular type of SMP, which require light as a stimulus to operate. LAP-based actuators are characterized by their rapid response times. Using only the variation of light frequency or its intensity, LAPs can be controlled remotely without any physical contact. The development of LAPs whose shape can be changed by the application of a specific frequency of light have been reported. The polymer retains its shape after removal of the activating light. In order to change the polymer back to its original shape, a light stimulus of a different frequency has to be applied to the polymer.

Actuator Characteristics:

The correct choice of actuators is necessary for the long-term sustenance and continuity of operations, as well as for increasing the lifetime of the actuators themselves. A set of four characteristics can define all actuators.

- **Weight:** The physical weight of actuators limits its application scope. For example, the use of heavier actuators is generally preferred for industrial applications and applications requiring no mobility of the IoT deployment. In contrast, lightweight actuators typically find common usage in portable systems in vehicles, drones, and home IoT applications. It is to be noted that this is not always true. Heavier actuators also have selective usage in mobile systems, for example, landing gears and engine motors in aircraft.

- **Power Rating:** This helps in deciding the nature of the application with which an actuator can be associated. The power rating defines the minimum and maximum operating power an actuator can safely withstand without damage to itself. Generally, it is indicated as the power-to-weight ratio for actuators. For example, smaller servo motors used in hobby projects typically have a maximum rating of 5 VDC, 500 mA, which is suitable for an operations-driven battery-based power source. Exceeding this limit might affect the performance of the actuator and may cause burnout of the motor. In contrast to this, servo motors in larger applications have a rating of 460 VAC, 2.5 A, which requires standalone power supply systems for operations. It is to be noted that actuators with still higher ratings are available and vary according to application requirements.

- **Torque to Weight Ratio:** The ratio of torque to the weight of the moving part of an instrument/device is referred to as its torque/weight ratio. This indicates the sensitivity of the actuator. Higher is the weight of the moving part; lower will be its torque to weight ratio for a given power.

- **Stiffness and Compliance:** The resistance of a material against deformation is known as its stiffness, whereas compliance of a material is the opposite of stiffness. Stiffness can be directly related to the modulus of elasticity of that material. Stiff systems are considered more accurate than compliant systems as they have a faster response

to the change in load applied to it. For example, hydraulic systems are considered as stiff and non-compliant, whereas pneumatic systems are considered as compliant.
